

DATA 505: Homework Quiz 2

Name: _____

2025-09-16

In R, suppose we have a object `penguins` with the following structure.

```
'data.frame':  344 obs. of  8 variables:
 $ species      : chr  "Adelie" "Adelie" "Adelie" "Adelie" ...
 $ island       : chr  "Torgersen" "Torgersen" "Torgersen" "Torgersen" ...
 $ bill_length_mm : num  39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ bill_depth_mm : num  18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length_mm: int  181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass_g   : int  3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex          : chr  "male" "female" "female" NA ...
 $ year         : int  2007 2007 2007 2007 2007 2007 2007 2007 2007 2007 ...
```

Question 1 (3 points)

Which of the following R operations would result in coercing the `species` variable to a **factor** object, and storing the result in the `species` column of the data frame? (*Circle the correct answer.*)

- A. `penguins.species <- penguins.species.as_factor()`
- B. `as_factor(penguins[["species"]])`
- C. `penguins[["species"]] <- as.factor(penguins$species)`
- D. There is no need for converting: `species` is already a factor object.

Question 2 (3 points)

Which of the following R operations would result in subsetting the data frame, keeping only those rows with non-missing body mass (`body_mass_g`), and only the `species` and `island` columns? (*Circle the correct answer.*)

- A. `penguins[!is.na(penguins$body_mass_g), c("species", "island")]`
- B. `penguins.[!is.na(body_mass_g); species, island]`
- C. `penguins[[!is.na("body_mass_g")]][c("species", "island")]`
- D. `penguins.drop_na("body_mass_g").select(["species", "island"])`

Question 3 (4 points)

```
summary(penguins$body_mass_g)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
2700	3550	4050	4202	4750	6300	2

Based on the output above, what is the interquartile range (IQR) of the body mass (`body_mass_g`)? (*Circle the correct answer.*)

- A. 4050 (the median)
- B. 4202 (the mean)
- C. 3600 ($6300 - 2700$)
- D. 1200 ($4750 - 3550$)